

JKBOSE 2026

CLASS 11TH

THE LIVING WORLD

ONE SHOT

BOTANY

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Topics To Covered

- 1** The Living World (Complete Chapter)

WHAT IS LIVING?

≡ 05 criteria to describe living organisms:

- a. Growth
- b. Reproduction
- c. Metabolism
- d. Cellular organisation
- e. Consciousness.

* Defining Property:

≡ Property +nt in all living organisms without any exception

* Characteristic property:

≡ Character of living organisms.

≡ It may have exception.

≡ It can also be seen in non-livings.

Note: * Growth
Reproduction } Characteristic Properties

* Metabolism
Cellular organisation
Consciousness } Defining properties.

GROWTH

≡ irreversible permanent increase in

- Weight
- Height
- Mass
- no. of cells.

≡ Growth

- a. Increase in mass
- b. Increase in no. of cells.

Twin characteristics of growth.



increase in
Size/mass



→ Binary Fission

increase in no.



Amoeba (unicellular)

≡ In plants — growth occurs continuously throughout their life span because of presence of meristematic tissue (Indeterminate growth).

≡ In Animals — growth is seen only upto certain stage. (Determinate growth).
 ↳ However cell division occurs in certain tissues to replace lost cells.

∴ Growth cannot be taken as defining property.

REASON:

∴ Growth in living organisms is protoplasmic i.e., occurs from inside (Intrinsic).

∴ Growth also occurs in non-livings like mountains & sand dunes due to accumulation of dust. (Extrinsic).

* Growth $\begin{cases} \text{Intrinsic in livings.} \\ \text{Extrinsic in non-livings.} \end{cases}$

NOTE: Growth in livings — involves $\begin{cases} \text{Synthesis of living materials ex. pm, cytoplasm etc.} \\ \text{and} \\ \text{Synthesis of dead materials ex. cell wall.} \end{cases}$

REPRODUCTION

≡ Biological process which involves production of new individuals.

≡ Two types :

1. Sexual reproduction : preferred by higher organisms.

- involves — gametogenesis , Fertilisation.
- Slow process.

2. Asexual reproduction : preferred by lower organisms.

- Gametogenesis and Fertilisation — Absent.
- Rapid Process.
- Various Types:

Binary Fission

Ex. Unicellular forms
Amoeba, Bacteria,
Paramecium

FRAGMENTATION

Ex. Spirogyra, Ulothrix,
• Rhizopus
• protonema of mosses

BUDDING

Ex. YEAST,
Hydra

SPORULATION

• By asexual spores.
Ex. Algae,
Fungi

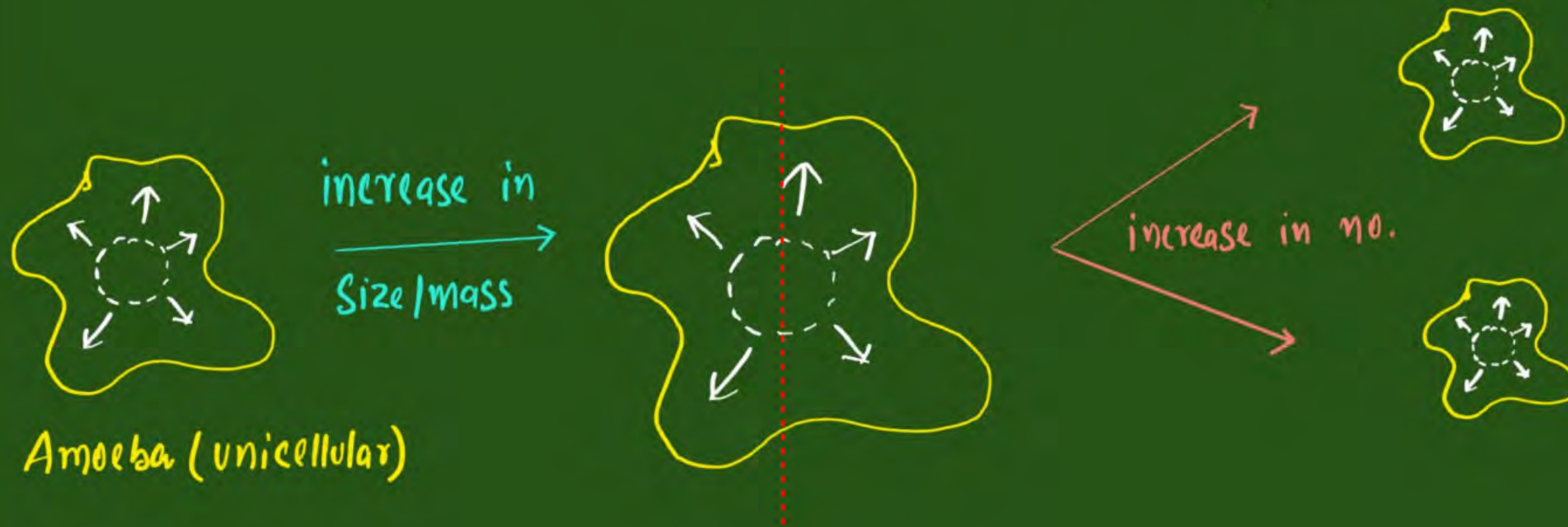
TRUE REGENERATION

Ex. Planaria
(Flat worm).

≡ Reproduction — absent in non-livings.

imp ≡ In case of unicellular organisms (eg. Amoeba) — Reproduction is increase in no. of cells.

⇓
which implies that in unicellular organisms, growth and reproduction are same/synonyms (i.e, mutually inclusive events i.e, increase in no. of cells).

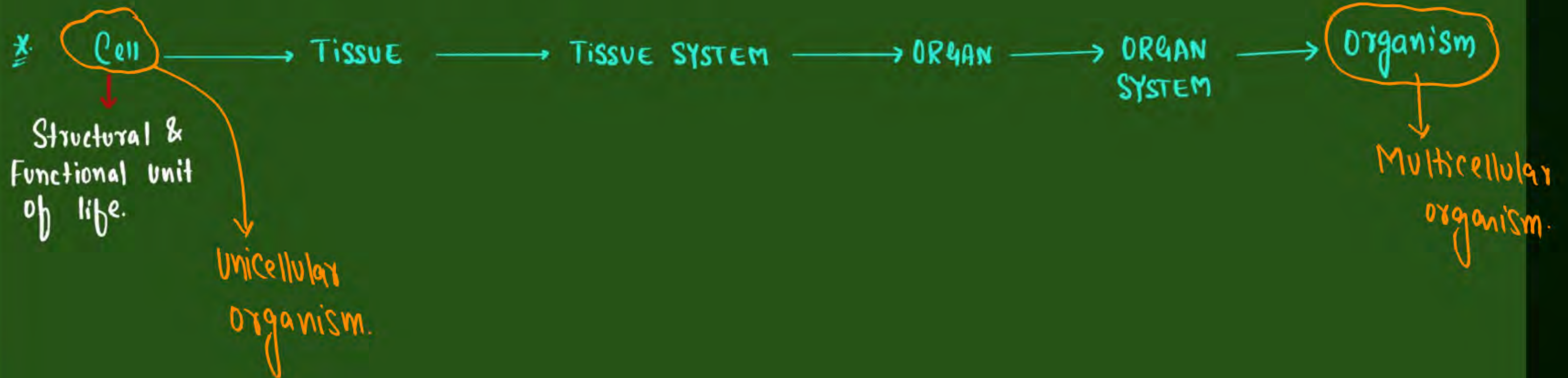


≠ Reproduction is not a defining property — because there are many living organisms which cannot reproduce like — Mules
 Infertile human couples.
 Worker bees (Honey bees).

CELLULAR ORGANISATION

≡ All living organisms are composed of cells.

≡
 → Unicellular organisms — Single celled (capable of independent existence and performs all essential functions of life).
 → Multicellular organisms — Many celled.



≡ Cellular organisation — is a defining property because all living organisms possess cellular organisation without any exception.

METABOLISM

≡ Sum total of all the chemical rxns. occurring in a body is termed as metabolism.

≡ Metabolism → Anabolism: • Build-up rxns.
• Ex. photosynthesis $6\text{CO}_2 + 12\text{H}_2\text{O} \longrightarrow \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{H}_2\text{O} + 6\text{O}_2$.

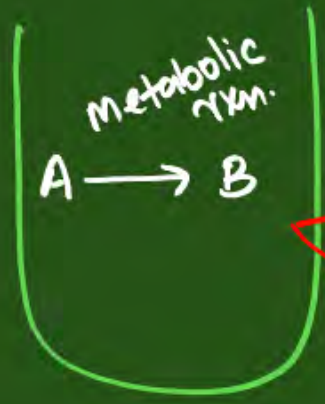
→ Catabolism: • Break down rxns.
• Ex. Respiration $\text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2 \longrightarrow 6\text{CO}_2 + 6\text{H}_2\text{O} + \text{ATP (Energy)}$

≡ Metabolic rxns. — are driven by biological enzymes which are only +nt inside the living organisms.

i.e, why metabolic rxns. takes place inside living organisms (inside cell).

Q. Can a metabolic rxn. takes place outside the body of living organisms eg. in test tube?

Ans. Yes — But Enzymes should be provided.



living organism (x)
Non-living (x)

∴ Isolated metabolic rxns. are neither living nor non-living but surely these are living rxns.

∴ Metabolism — is a defining property of living organisms
Because no non-living performs metabolism
and all living organisms performs metabolism.

CONSCIOUSNESS

≡ Defined as ability to sense the surrounding environments and respond to the environmental stimuli.

i.e, $\text{Consciousness} = \text{Perception of Stimulus} + \text{Response}$

≡ Stimulus can be

- Biological (ex. Mosquito bite)
- Chemical (ex. Perfumes etc.)
- Physical (ex. light, temp. etc.)

≡ Consciousness — is the defining property of living organisms because all living organisms respond to environmental stimuli.

≡ Consciousness — Seen in plants ex. perform flowering in particular season (photoperiodism).

≡ Very high level of Consciousness is found in humans because of well developed nervous system and ultimate level of communication skill which is called as Self-Consciousness.

≡ Self-Consciousness — Not a defining property.

≡ Human being is the only organism who is aware of himself i.e., Self Conscious (can plan future, think of past, aware of existing world).

NCERT

≡ In case of humans - it is difficult to define living state

⇓

"Coma patients"

↳ ≡ Patient is brain dead. *

≡ These patients have consciousness becoz they respond to chemicals entering into their body.

≡ No self consciousness becoz they have lost the coordination of organs of different body parts.

NOTE:

* All living organisms have originated from a common ancestor kld "progenote"

⊗ All living organisms share common genetic material (i.e., DNA) but to varying degrees.

BIODIVERSITY

≡ Biodiversity

- "Bio" — means "life"
- "diversity" — means "forms / variety"

≡ Biodiversity — refers to number and types of living organisms +nt on planet Earth.

≡ 1.7 to 1.8 Million Sps. — Scientifically named and classified } **Taxonomically Known Sps.**

- 1.2 million — include Animal Sps.
- 0.5 million — include Plant Sps.

≡ Among Animals — largest group is Arthropoda (insects) — > 10,25,000 Sps.

≡ Among Plants — largest group is Angiosperms (Flowering plants) — > 80%.

NEED FOR CLASSIFICATION:



≡ There are millions of plant and animal sps.



Varying greatly in their form, structure and complexity.

≡ It is very difficult to study all of them individually.

≡ In order to make study easy — organisms have been divided into different categories/ranks on the basis of similarities and differences.

Q. What is classification?

Ans.

H/w.

TAXONOMY

≡ Taxonomy — derived from two Greek words $\begin{cases} \text{"Taxis"} - \text{Arrangement.} \\ \text{"Nomos"} - \text{laws / rules.} \end{cases}$

≡ Taxonomy — branch of Science which deals with process of classification of living organisms by following rules/laws.

≡ Term taxonomy — given by "A.P deCandolle" in his book — "Theory of Elementary Botany"

≡ Carolus Linnaeus — regarded as "Father of taxonomy."

≡ Taxonomy — Two types:

a. Classical Taxonomy :-

≡ involves use of morphology only (i.e, External characters).

b. Modern Taxonomy :-

≡ involves use of —

morphological characters	(External)
Anatomical characters	(internal)
Cytological	" (cell)
Developmental	"
Ecological	" (e.g. Habitat etc.).

STEPS INVOLVED IN TAXONOMY

a. CHARACTERISATION :-

≡ Organism is described according to its all morphological, anatomical, physiological and behavioural properties.

OR
Account of Characters.

b. Identification :-

≡ Organism is recognised from other already known organisms and is assigned to a particular taxonomic group.

OR
Comparison with other known groups.

c. Classification :-

≡ Organism is grouped into convenient categories on the basis of some easily observable characters.

OR
Assign a group.

d. Nomenclature :-

≡ Naming of an organism according to established rules.

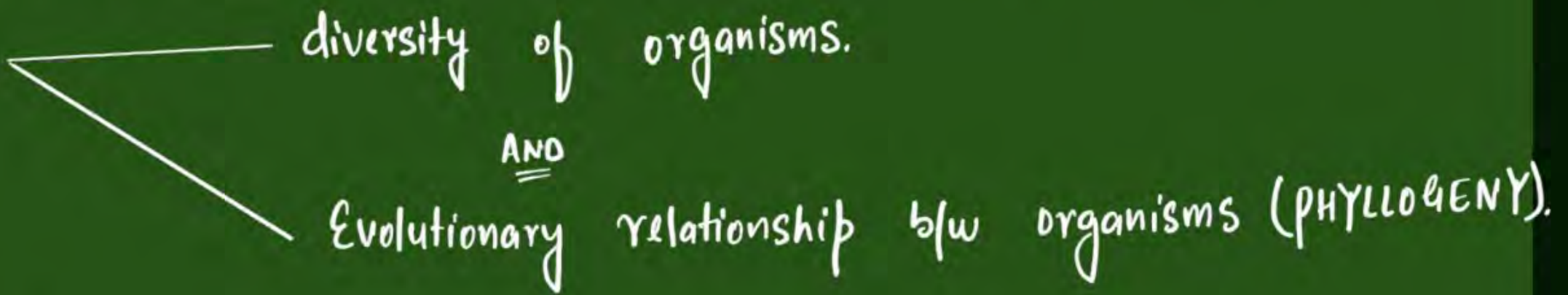
OR
Assign a Scientific name.

SYSTEMATICS



≡ Word Systematics is derived from latin word — "Systema"
↓
Which means systematic arrangement of living organisms.

≡ Term "Systematics" — used by Carolous Linneaus in his book "Systema Naturae"

≡ Systematics deals with  diversity of organisms.
AND
Evolutionary relationship b/w organisms (PHYLOGENY).

≡ In Modern times,

Systematics includes

TAXONOMY ✓

AND

PHYLOGENY ✓ (Evolutionary relationship b/w organisms).

⊛ Systematics = Taxonomy + Phylogeny

NOMENCLATURE

- ≡ PROCESS of assigning a name to an organism.
- ≡ Two types of Names have been given to organisms.

a. Common / Vernacular name / local name / regional names:

- ≡ These names are given to organisms in a particular language & region of world.
- ≡ These names are full of ambiguity. (Confusion).

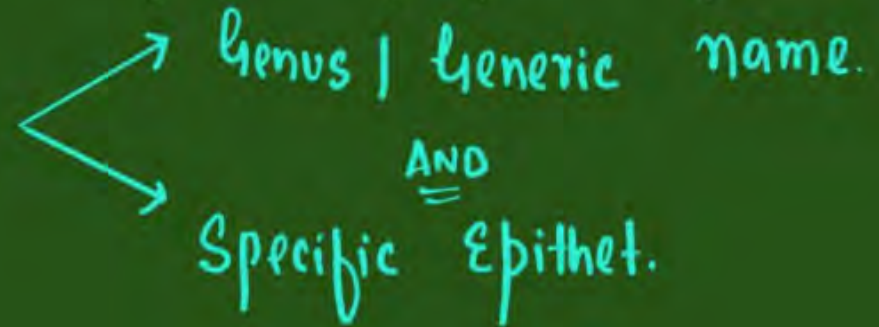
b. Scientific names / technical names: (Biological names).

- ≡ These names are given by biologists and is understood to represent a particular organism in every part of world.
- ≡ No ambiguity.

BINOMIAL NOMENCLATURE

≡ Given by Carolus Linnaeus.

≡ The system of providing a name with two components is k/d Binomial nomenclature.



Ex. Solanum tuberosum (potato)

↓ ↓

Genus / Generic name Specific Epithet

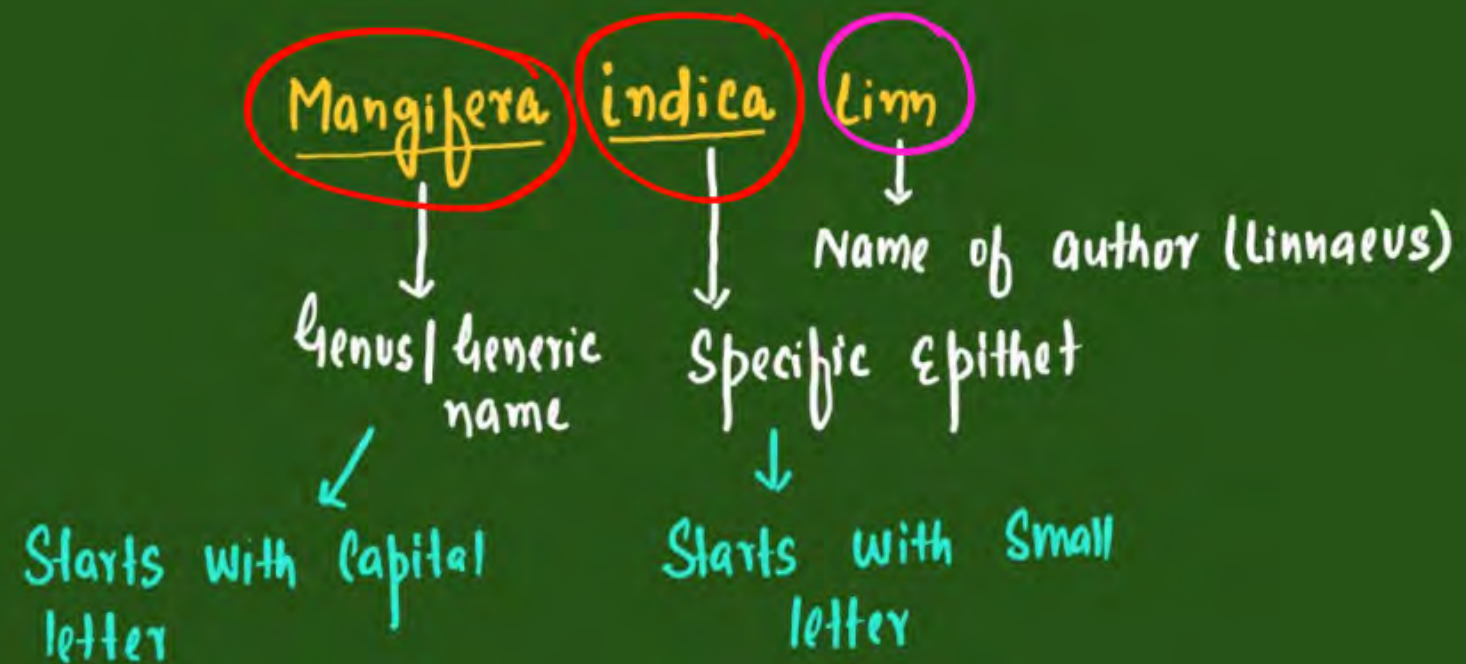
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Note: Species name is the complete name of a species (Genus + Specific Epithet).

≡ Carolus Linnaeus — used binomial nomenclature in his book "Species plantarum."

RULES OF BINOMIAL NOMENCLATURE

Ex. Scientific name of Mango is :



1. Biological names are generally taken from Latin language irrespective of their origin (or are latinized).

2. Biological names — consist of two components/words $\left\{ \begin{array}{l} \text{Generic name} \\ \text{AND} \\ \text{Specific epithet} \end{array} \right.$

Ex. Mangifera indica (Mango)

Solanum tuberosum (potato)

3. Scientific names — printed in italics or underlined separately when handwritten to indicate their latin origin.

4. First letter of genus — Starts with Capital letter.
First letter of Specific epithet — Starts with Small letter.

5. Name of Author / discoverer — is written after specific epithet in abbreviated form. It is not underlined nor italicized (i.e. written in simple alphabets).

Ex. Mangifera indica Linn
Homo sapiens Linn

→ "Carolus Linneous"

CODES OF NOMENCLATURE

≡ Agreed Convention for naming biological groups.

ex. ICBN — International Code for Botanical nomenclature. (plants)
 ICZN — International Code of Zoological nomenclature. (Animals)

COMMON NAMES

1. POTATO
2. Brinjal
3. Makoi / Black night shade
4. LION
5. LEOPARD
6. Tiger
7. Dog
8. wolf
9. Cat
10. Man
11. Housefly
12. Mango
13. wheat

SCIENTIFIC NAMES

Solanum tuberosum
Solanum melongena
Solanum nigrum
Panthera leo ✓
Panthera pardus ✓
Panthera tigris ✓
Canis familiaris
Canis lupus
Felis catus
Homo sapiens
Musca domestica
Mangifera indica
Triticum aestivum.



TAXONOMIC CATEGORIES

≡ Classification — is not a single step process.

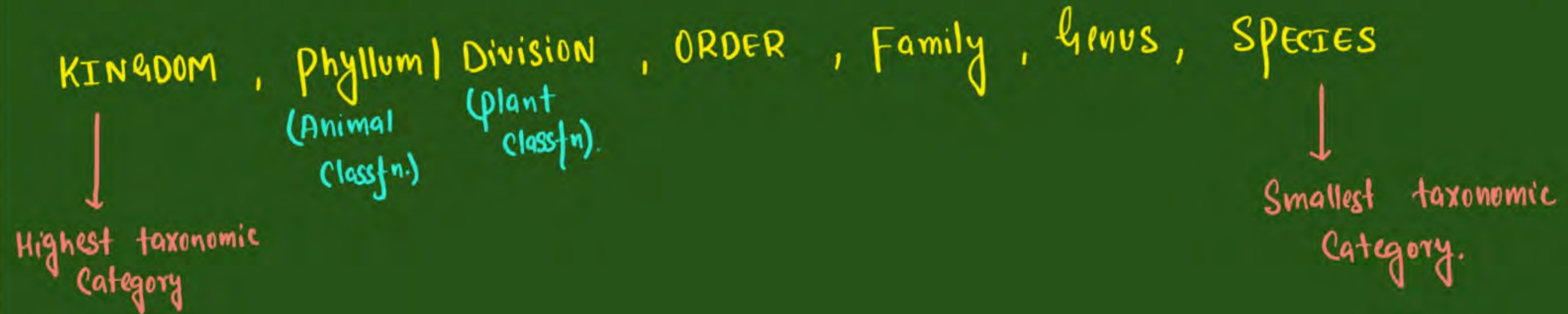
≡ It is multistep process and thus involves hierarchy of steps in which each step represents a rank / category.

x. Taxon — unit of classification. (plural. Taxa).

NOTE:- Thin line of difference b/w Taxon and Category.

<u>TAXON</u>		<u>CATEGORY</u>
It represents living entity.	→	It represents rank of organism.
Ex. <u>tuberosum</u>	→	Species
<u>Solanum</u>	→	"Genus"

* There are 07 main taxonomic categories which are obligate / essential i.e, why they are strictly used at the time of any plant classification.

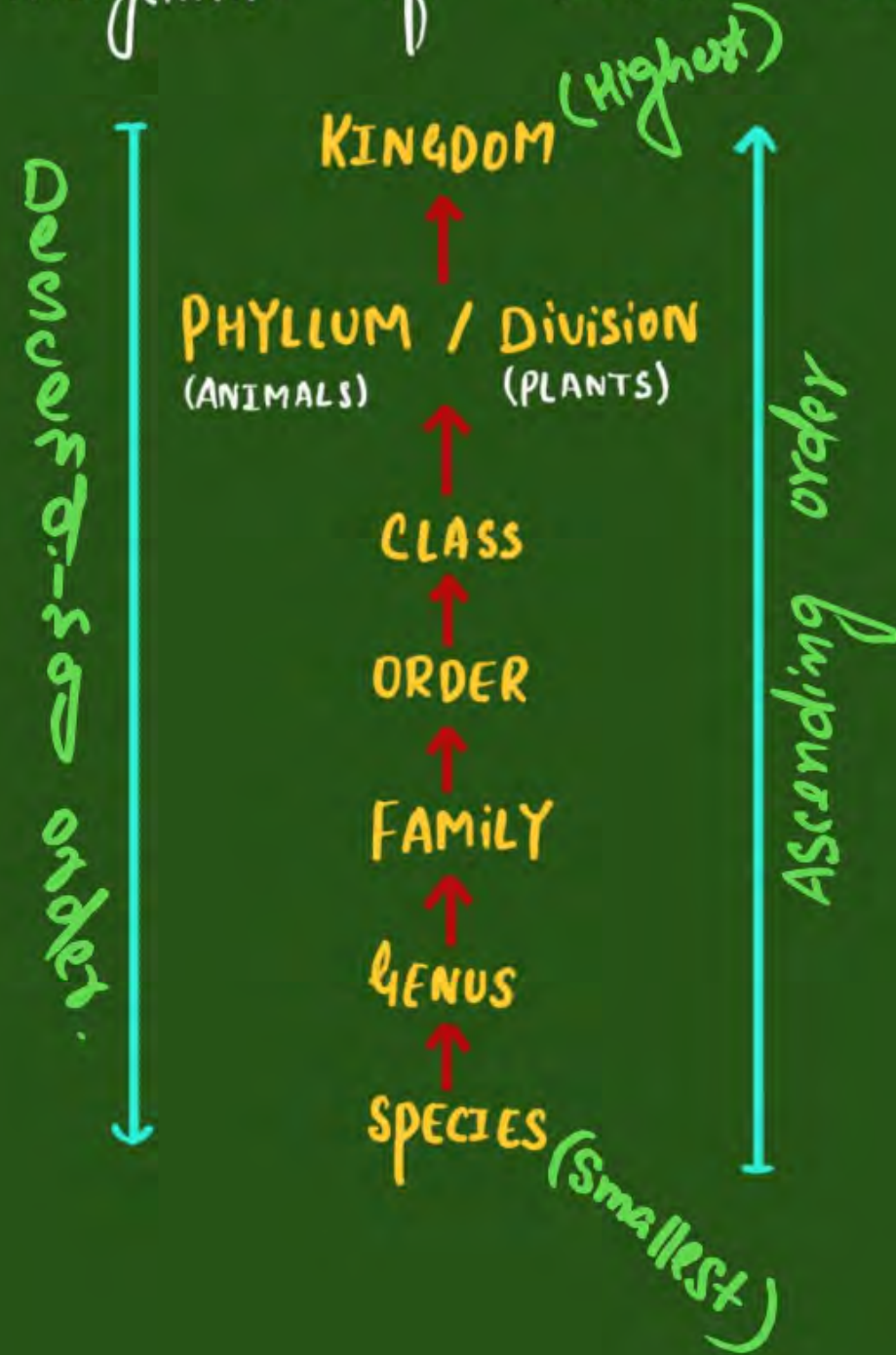


* These taxonomic categories are distinct morphological entities and not merely

مرز

TAXONOMIC HIERARCHY

≡ Arrangement of taxonomic categories — either in ascending or descending order.



NOTE: AS WE MOVE FROM SPECIES TO KINGDOM:

- ①. No. of similar characters i.e, specificity — decreases.
- ②. No. of general characters — increases
- ③. Species diversity — increases
- ④. Complexity of classification — increases.

SPECIES

≡ Real basic unit of classification.

* NOTE: Unit of classification — TAXON.

≡ Species — group of individual organisms with fundamental similarities as a species.

↓
defined by "Biological Species Concept" — proposed by "Ernst Mayr"

⇓
As per this Concept, species is a group of organisms which:



- a. Closely related (Structurally and functionally) ✓
- b. Can interbreed freely in nature & produce fertile offsprings.
- c. are reproductively isolated from other sps. (Main basis of biological sps. concept.)

NOTE :

1. Biological Species Concept is not applicable everywhere.

Ex. Mule (Sterile) = ♂ Donkey x ♀ horse

2. Ernst Mayr
(1904 - 2004)

→ Kld as "Darwin of 20th Century"

→ Evolutionary Biologist.

→ Awarded by 03 prizes (AKA Triple Crown of biology)

✓ a. Balzon Prize (1983)

✓ b. International prize for Biology (1994)

✓ c. Crafoord prize (1999)

GENUS

≡ Group of related Species.

≡ Genus has more characters in common in comparison to species of other genera.

Genus ①

Sps. a } Max.
Sps. b } Common Chrs.

Genus ②

Sps. a } Max. Common
Sps. b } Characters.

less common characters.

≡ Each genus — may have one/more than one specific epithet.

Ex. 1. Solanum
(Genus)

- tuberosum
- melongena
- nigrum

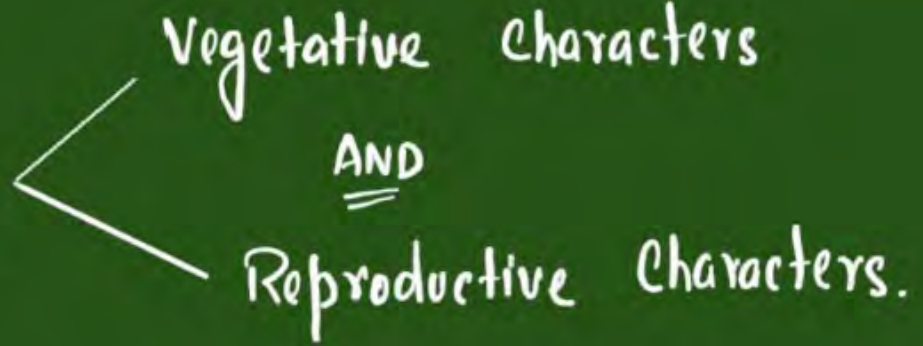
2. Panthera
(Genus)

- pardus
- leo
- tigris

3. Felis — Include Cats
(Genus)
Ex. Felis catus

FAMILY

≡ group of related genera.

≡ Plant families are characterised on the basis of  Vegetative characters AND Reproductive Characters.

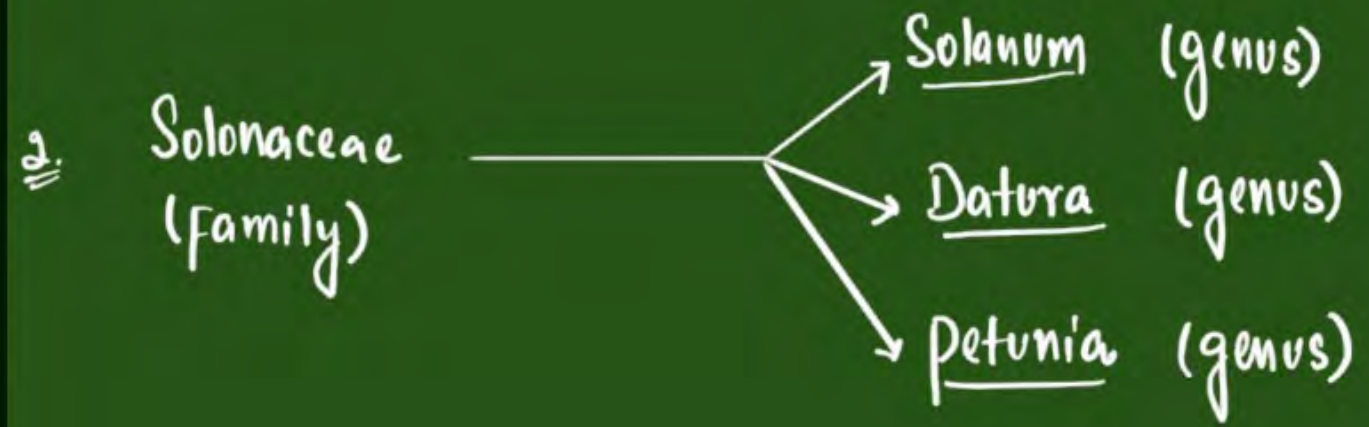
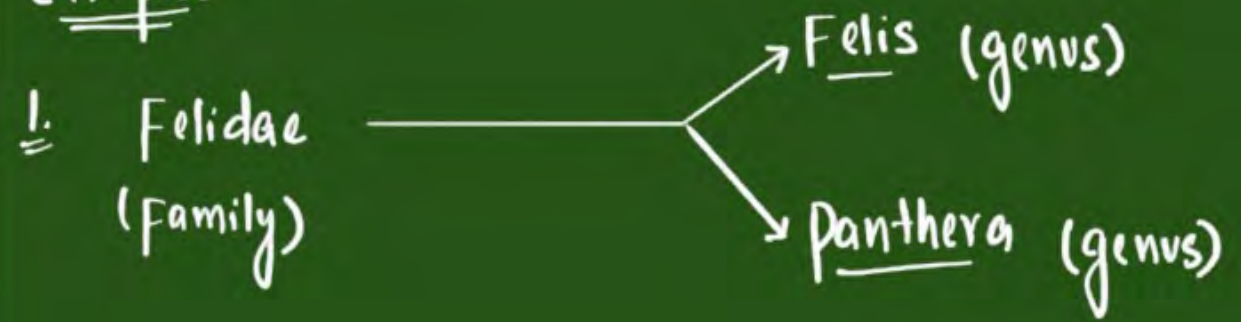
≡ In animal families, suffix -idae is used mostly

ex. Felidae (Cat Family)
Canidae (Dog Family)
Hominidae (Human Family)
Muscidae (Housefly Family)

≡ In plant families, suffix -aceae is used mostly.

ex. Solanaceae (Potato Family)
Liliaceae (Lily Family)
Fabaceae (Bean Family)
Poaceae (Grass Family)

Examples



ORDER

≡ group of related Families.

≡ Some animal orders are : Primata, Diptera, Rodentia, Carnivora.

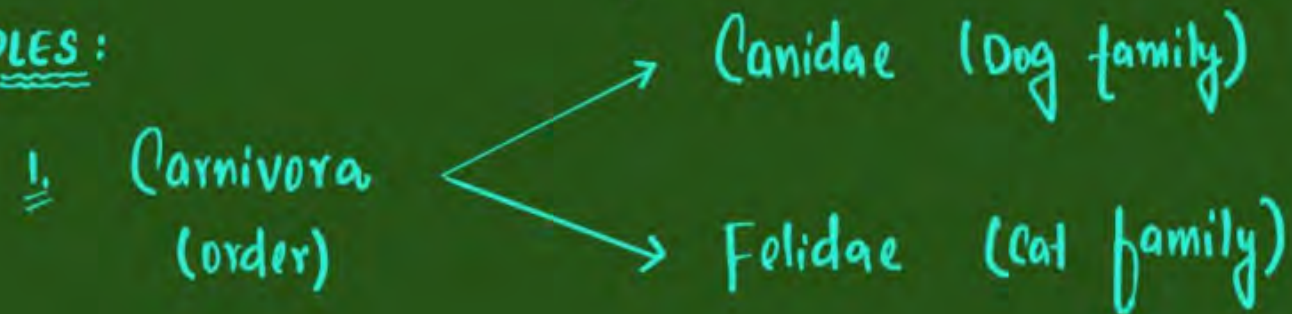
≡ In plant orders, Suffix "-ales" is used

Ex. Sapindales, Poales, Polymoniales.

≡ Plant orders — are formed on the basis of floral characters :

"Flower characters"

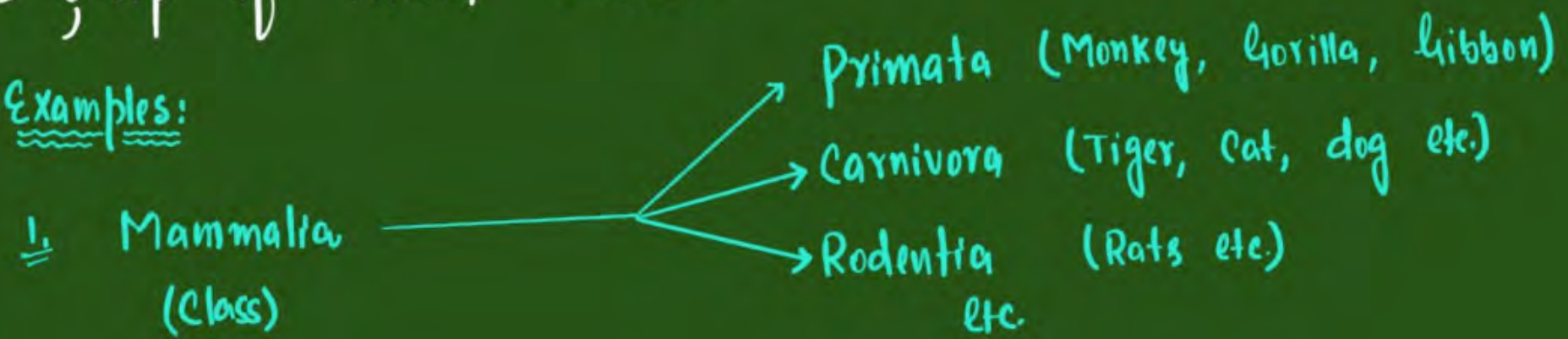
EXAMPLES :



CLASS

≡ group of related orders.

Examples:



PHYLLUM/DIVISION

≡ phylum — used in animal classification.
 Division — used in plant classification.

≡ Group of related classes.

Examples:



KINGDOM

≡ Group of related Divisions / Phyla.
 ≡ highest taxonomic Category.

Example :

1. Plant Kingdom — includes All plants.
2. Animal Kingdom — includes All Animals.

TABLE 1.1 Organisms with their Taxonomic Categories

Common Name	Biological Name	Genus	Family	Order	Class	Phylum/ Division
Man	<i>Homo sapiens</i>	<i>Homo</i>	Hominidae	Primata	Mammalia	Chordata
Housefly	<i>Musca domestica</i>	<i>Musca</i>	Muscidae	Diptera	Insecta	Arthropoda
Mango	<i>Mangifera indica</i>	<i>Mangifera</i>	Anacardiaceae	Sapindales	Dicotyledonae	Angiospermae
Wheat	<i>Triticum aestivum</i>	<i>Triticum</i>	Poaceae	Poales	Mono: cotyledonae	Angiospermae
Wheat	<i>Triticum aestivum</i>	<i>Triticum</i>	Poaceae	Poales	Monocotyledone	Angiospermae

HERBARIUM

Herbarium
Sheet

Herbarium



≡ It is a store house of "dead plant specimens only".

≡ In Herbarium, plant specimens are —

- Collected ✓
- Dried ✓
- pressed ✓
- and preserved on sheets.

≡ Herbarium — act as a quick referral system for classification. ✓

≡ Herbarium Sheets — are arranged according to universally accepted system of classification.

≡ Herbarium Sheets — Carry a label (on right hand bottom) providing information about

- a. Date and place of Collection. ✓
- b. English name, Common name, Botanical name. ✓
- c. Family ✓
- d. Collector's name. ✓

≡ International Size of herbarium Sheet = 29 x 41.5 cm.

Thank
You